

Year 10 Engineering Technology - Water Control System

20-week scope and sequence for the Garden Watering / Water Control System unit

Unit Overview

Students design, construct, test and evaluate a small water-control system for a garden watering context. The unit uses the NESA Engineering Technology 7-10 Control systems focus area and draws on electronic and hydraulic control contexts.

This document spreads the unit across a full 20-week semester. Timing can still be adjusted around excursions, assessment windows and school events.

Stage 5 Outcomes

Outcome	Official outcome wording used for this unit
EGT5-SAF-01	applies risk management and safe work practices in engineering contexts
EGT5-COM-01	communicates ideas, concepts and solutions for engineering practice
EGT5-EVL-01	investigates and evaluates engineering systems and solutions
EGT5-IVT-01	explains engineering practices and the influence of technologies used in engineering industries
EGT5-GRP-01	develops and applies technical graphics in engineering contexts
EGT5-MEA-01	applies mechanical analysis and practical testing to investigate engineering concepts
EGT5-USE-01	selects and applies materials for engineering projects
EGT5-ENV-01	analyses relationships between engineering design, production and sustainability

Content Alignment

Content area	Where it appears across the 20 weeks
Input, control and output principles	Weeks 2, 6 and 7 through IPO diagrams, feedback comparison and algorithm/flowchart development.
Electronic control systems	Weeks 5, 8, 14 and 15 through sensor/switch, relay/controller and output testing.

Hydraulic/water-control systems	Weeks 5, 13, 15 and 16 through reservoir, tubing, pump/valve and leak-testing work.
Safe work practices	Weeks 1, 3, 8, 11, 13-16 through wet/dry zones, low-voltage checks and teacher sign-offs.
Engineering process	Weeks 4-18 through brief analysis, design, construction, integration, testing and improvement.
Testing and troubleshooting	Weeks 12, 16-18 through repeated tests, response time, water delivered and fault logs.
Technical communication	Weeks 6-12 and 19-20 through sketches, diagrams, tables, folio/report and evaluation.
Sustainability	Weeks 4, 18 and 19 through water efficiency, material choice, repairability and reuse/recycling.

20-Week Scope and Sequence

Week	Focus	Learning experiences	Target outcomes	Evidence
1	Unit launch, WHS and control-system context	Project brief; everyday irrigation and automation; low-voltage safety; wet and dry work zones.	EGT5-SAF-01, EGT5-IVT-01, EGT5-COM-01	Safety notes, first project brief response and hazard list.
2	Inputs, outputs and feedback	Input-process-output language; open-loop and closed-loop control; moisture, rain, switch and timer inputs.	EGT5-EVL-01, EGT5-COM-01, EGT5-IVT-01	IPO diagram, vocabulary notes and feedback explanation.
3	Safe testing setup	Workshop layout; spill control; dry electronics zone; teacher checkpoints; safe tool and component handling.	EGT5-SAF-01, EGT5-COM-01	Annotated safe test setup and risk-control checklist.
4	User need and success criteria	Garden watering problem; user needs; water efficiency; measurable success criteria and folio setup.	EGT5-COM-01, EGT5-ENV-01	User-need statement, success criteria and folio structure.
5	Sensors and actuators	Moisture/rain sensors, switches, solenoids, pumps, indicators, relays and controllers.	EGT5-USE-01, EGT5-MEA-01, EGT5-EVL-01	Component comparison table and annotated component sketch.
6	IPO diagrams and system	System boundaries; block diagrams; feedback	EGT5-COM-01, EGT5-	Block diagram and glossary entries.

	vocabulary	examples; control-system language used in context.	IVT-01	
7	Algorithms and control logic	Flowchart or pseudocode; watering decisions; trigger thresholds; timing and reset logic.	EGT5-GRP-01, EGT5-COM-01	Algorithm or flowchart with teacher feedback.
8	Component research and safe test sequence	Circuit options; pump or valve options; test order; data table design; fault-log setup.	EGT5-EVL-01, EGT5-SAF-01	Research notes, test sequence and draft fault log.
9	Concept sketching	Reservoir, tubing, plant area, sensor position, controller location and dry electronics zone.	EGT5-GRP-01, EGT5-COM-01	Two concept sketches with annotations.
10	Final design selection	Compare concepts; choose final layout; labelled drawing; circuit/wiring plan; irrigation layout.	EGT5-GRP-01, EGT5-EVL-01	Selected concept, justification and labelled final design.
11	Build planning and approval	Materials/component list; tools; step-by-step build sequence; updated risk controls; teacher approval.	EGT5-SAF-01, EGT5-COM-01	Build plan, component list and approval checkpoint.
12	Testing preparation	Testing tables; performance measures; response time; water delivered; folio evidence checklist.	EGT5-MEA-01, EGT5-EVL-01	Test tables, evidence checklist and prototype setup plan.
13	Water path construction	Build base, reservoir, tubing route and outlet; leak testing with power disconnected.	EGT5-SAF-01, EGT5-USE-01, EGT5-MEA-01	Build log, photos and leak-test record.
14	Input/control subsystem construction	Build and check input or switch circuit; wiring checks; test triggering without water flow.	EGT5-SAF-01, EGT5-MEA-01	Input test data, wiring photo or diagram and troubleshooting notes.
15	Output subsystem testing	Connect pump or valve; short subsystem tests; water delivery check; teacher checkpoint.	EGT5-SAF-01, EGT5-EVL-01	Output test data and teacher check record.
16	System integration and fault-finding	Connect subsystems; diagnose leaks, wiring faults, unreliable triggering or poor delivery.	EGT5-EVL-01, EGT5-MEA-01, EGT5-COM-01	Full-system test table and fault log.
17	Reliability testing	Repeated test cycles; response time; water delivered; consistency and failure patterns.	EGT5-EVL-01, EGT5-MEA-01	Repeated-test data and reliability notes.
18	Controlled improvements	Modify one variable at a time; retest; compare	EGT5-EVL-01, EGT5-	Improvement record and

	and retesting	before and after results.	COM-01	comparison table.
19	Sustainability and final performance review	Water use, material choice, cost, repairability, recycling/reuse and final performance test.	EGT5-ENV-01, EGT5-EVL-01	Sustainability reflection and final test evidence.
20	Folio completion, evaluation and reflection	Engineering report structure; peer review; final evaluation; submission and class reflection.	EGT5-COM-01, EGT5-EVL-01, EGT5-IVT-01	Completed folio/report and project reflection.

Assessment and Evidence

- Prototype: a functioning water-control system that demonstrates input, control/process and output.
- Folio/report: brief analysis, research, IPO/block diagram, sketches, final design, algorithm, risk controls, build log, test data, troubleshooting and evaluation.
- Teacher observation: safe work practices, tool/component use, testing behaviour and collaboration.
- Evaluation: reliability, efficiency, safety, sustainability, water use and justified improvements.